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Amniotic Fluid Embolism

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Continuing Education Activity

Amniotic fluid embolism (AFE) is a rare but serious condition with high mortality and morbidity rates and is the second leading cause of peripartum maternal death. It occurs when amniotic fluid enters the maternal bloodstream. This article discusses the clinical presentation of AFE, providing a detailed description of its pathophysiology and the complex evaluation and treatment methods involved. Cardiovascular collapse often serves as the initial clinical indicator of AFE. Next, activation of the coagulation cascade and fibrinolytic systems can lead to disseminated intravascular coagulation. Diagnosing AFE relies on clinical judgment, as there is no specific test for confirmation.

This activity equips healthcare professionals with the knowledge needed to navigate the complexities of AFE, providing a platform to enhance clinical acumen, improve diagnostic precision, and optimize collaborative care, ultimately contributing to improved outcomes in this critical obstetric emergency. Participants will gain insights into the global epidemiology and regional variations, focusing on the alarming rates in the United States. Clinicians will attain a comprehensive understanding of the condition's distinct pathophysiology, initial clinical indicators, and evolving research and guidelines to continually improve clinical competence.

Objectives:

- Identify the various risk factors for amniotic fluid embolism.
- Assess the triggering pulmonary pathology initiated by the inflammation associated with amniotic fluid embolism.
- Implement evidence-based evaluation and treatment methods for amniotic fluid embolism, ensuring prompt and effective responses to this critical obstetric emergency.
- Collaborate in improving care coordination among an interprofessional team to enhance the delivery of care for a patient affected by an amniotic fluid embolism.

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Introduction

Amniotic fluid embolism (AFE) is a critical obstetric emergency marked by sudden cardiorespiratory collapse and disseminated intravascular coagulation (DIC). Often referred to as the anaphylactoid syndrome of pregnancy, AFE is distinct from a direct embolism or the presence of amniotic fluid itself.[1][2] The global occurrence of AFE ranges from 1 in 8000 to 1 in 80,000 deliveries, though the exact incidence remains unclear due to diagnostic and reporting inaccuracies.[2]

In the United States, AFE ranks as the second leading cause of peripartum maternal death and the primary cause of peripartum cardiac arrest. Its presentation is abrupt, typically involving sudden cardiorespiratory collapse, altered mental status, and subsequent severe coagulopathy.[1] Survivors often contend with serious cardiac, renal, neurologic, and pulmonary dysfunction. In the United States, AFE affects 2.2 to 7.7 per 100,000 deliveries, contributing to 7.5% of maternal deaths. In developing countries, the mortality rate ranges from 1.8 to 5.9 per 100,000 deliveries, compared to 0.5 to 1.7 per 100,000 deliveries in developed countries.[1]

The initial description of AFE dates back to 1941 when Steiner and Luschbaugh noted fetal cells in the maternal pulmonary circulation of women who died during labor.[1][3] Data from the National Amniotic Fluid Embolism Registry indicates that the condition shares similarities with anaphylaxis rather than a typical embolism. Notably, fetal tissue or amniotic fluid components are not consistently detected in women displaying signs and symptoms of AFE.[1] Traditionally, AFE diagnosis occurred postmortem, relying on the presence of fetal squamous cells in the maternal pulmonary artery blood.[4] However, as fetal squamous cells are also found in the circulation of laboring women who do not develop AFE, the diagnosis is exclusionary and based on clinical presentation after ruling out other causes of hemodynamic instability.[5]

Etiology

AFE remains unpredictable, with an elusive origin. Various factors contribute to its development, encompassing maternal age (especially older than 35 to 40 years), grand multiparity, male fetuses, early gestational age, cervical ripening, polyhydramnios, multiple gestation, gestational diabetes, operative delivery, manual placental extraction, regional disparities (western U.S.), Asian and Black races, asthma, illicit substance use, and trauma.[6] Induction of labor and conditions such as cerebrovascular disorders and cardiac disease heightens the risk of AFE, with strong associations observed in placenta previa, eclampsia, uterine rupture, fetal growth restriction, fetal death, placental abruption, maternal renal disease, peripartum cardiomyopathy, and postpartum hemorrhage.[6]

There is some conflicting evidence on the risk factors for AFE.[7] Many earlier studies concluded that cesarean delivery, particularly with a classical uterine incision, raised AFE risk. This concept has since been reversed, with cesarean section of any type found to be unrelated. Similarly, amniotomy was once thought to increase the risk of AFE but is now believed to be uncoupled. One population-based case-cohort study looked at 149 cases of AFE, 80 of which were fatal. They reported that a spontaneous vaginal delivery had 12 times the risk of AFE than cesarean delivery, and instrumental vaginal delivery was almost 3 times the risk of cesarean delivery. They concluded that cesarean section is a protective factor for a fatal AFE.[3] Amnioinfusion does correlate with a 3-fold increase in AFE risk, potentially due to increased uterine distension.[6] Notably, 66% of AFE patients reported prior allergies, aligning with the condition's secondary name, "anaphylactoid syndrome of pregnancy," exceeding the general population's atopy rate. Additionally, 8% of AFE-affected pregnancies stem from in vitro fertilization, surpassing the baseline IVF rate.[6]

Placenta accreta spectrum (PAS) disorder emerges as the anomaly most strongly linked to AFE, carrying a 10-fold increased risk.[6][8] The severity of PAS correlates with a higher AFE incidence.[6] Introducing amniotic fluid and fetal components into maternal circulation triggers intense pulmonary vasoconstriction and bronchoconstriction. These effects result not only from physical obstruction but primarily from the release of inflammatory cytokines reacting to foreign material.[8] These mediators activate coagulation and fibrinolytic pathways, culminating in a DIC syndrome.[6]

Epidemiology

The estimated incidence of AFE ranges from 1.9 to 6.1 per 100,000 births, though the exact prevalence remains uncertain due to inaccurate diagnosis and underreporting of nonfatal cases.[9][10] Notably, AFE emerged as the primary cause of death during parturition in Germany in 2011 and accounts for 24.3% of maternal deaths in Japan.[10] In Australia, AFE is recognized as the leading direct cause of maternal mortality, affecting between 1 in 8000 to 1 in 80,000 deliveries. The estimated incidence in the UK is 2 per 100,000 births, while in the United States, AFE occurs at approximately 7.7 per 100,000 births.[10]

A significant majority of AFE cases, around 70%, manifest during labor, with approximately 19% occurring during cesarean sections and 11% following vaginal deliveries. It is noteworthy that AFE can occur up to 48 hours after delivery.[10] Rare instances of AFE have been reported after pregnancy termination, amniocentesis, injection of hypertonic saline into the uterus for abortion induction, and in the first or second trimester of pregnancy.[5]

Pathophysiology

AFE is characterized by a disruption in the placental-amniotic interface, leading to the entry of amniotic fluid and fetal elements, such as hair, meconium, skin cells, and gut mucin, into the maternal circulation. It is essential to note

that the presence of squamous cells in the pulmonary circulation is no longer solely diagnostic for AFE, as clinical presentation plays a crucial role.[11][12][7]

Accompanying the entry of amniotic and fetal substances are tissue factors with procoagulant properties.[11] Activation of histamine, endothelin, and leukotrienes leads to physiologic changes that result in cardiovascular collapse.[13] Potential portals of entry include the placental attachment, cervical veins, or uterine surgical incisions. Once within the pulmonary arterial tree, this triggers a pathological maternal anaphylactoid immune response, releasing inflammatory mediators.[11]

The initial phase involves intense and transient pulmonary vasoconstriction, possibly coupled with bronchoconstriction. This results in acute pulmonary arterial obstruction, dilation of the right ventricle and right atrium, and significant tricuspid regurgitation. Subsequently, hypoxia and right heart failure ensue. There is a less common type of AFE reported, which presents with only the hemorrhage and DIC component without maternal hemodynamic instability.[4][11]

Subsequent to right ventricular enlargement, left ventricular function experiences significant depression due to myocardial ischemia stemming from hypoxia or coronary artery spasm. This ventricular enlargement causes the intraventricular septum to bow into the left ventricle, resulting in obstruction and systolic dysfunction. Consequently, pulmonary artery pressure rises, and cardiac output decreases.[7] Dysrhythmias such as ventricular fibrillation, asystole, and pulseless electrical activity have all been reported in association. Thus, survivors of this critical condition may face hypoxic brain injuries or multisystem organ failure.[14]

Sudden cardiovascular collapse is induced by hypoxemia and hypotension. The introduction of amniotic fluid and fetal elements triggers inflammatory mediators, including platelet-activating factor, tissue necrosis factor-alpha (TNF-alpha), interleukin 6, interleukin 1, phospholipase A2, endothelin, plasminogen activators, thromboplastins, and complement factors.[14] This activation sets off the coagulation cascade and fibrinolytic systems, resulting in a fibrinolytic form of DIC. Amniotic fluid in the maternal circulation activates platelet factor III, leading to platelet aggregation and activation of clotting factor Xa. Amniotic fluid and fetal elements may infiltrate the uterus, causing severe uterine atony and exacerbating hemorrhage. The superimposed pathologic activation of the coagulation and fibrinolytic pathways results in severe coagulopathy, observed in approximately 80% of AFE patients.[14] The decrease in coagulation factors may occur either immediately at the time of cardiopulmonary collapse or in a delayed manner. Bleeding can be severe, relentless, and fatal.

Autopsies on women succumbing to AFE reveal pulmonary edema, amniotic component emboli in the lungs, and alveolar hemorrhage. Additional findings may include myocardial infarction, acute renal failure from acute tubular necrosis, and cerebral infarctions.

Histopathology

Prevalence of Pulmonary Edema

- Pulmonary edema is a prevalent finding in 70% of postmortem examinations of individuals who succumb to AFE.
- This condition constitutes a crucial pathological feature, underscoring its significance in AFE cases.

Microscopic Presence of Amniotic Fluid Substances

- Despite the presence of amniotic fluid substances in the lungs, their microscopic identification can be challenging due to their minute size.
- Histological examinations may not always capture these tiny particles, potentially leading to underrecognition.[15]

Alveolar Hemorrhage

- In conjunction with pulmonary edema, alveolar hemorrhage is a common histological manifestation in the lungs of individuals affected by AFE.

- The observation of alveolar hemorrhage adds another layer to the pathological changes associated with this condition.

These clinical pearls shed light on the intricate pathological aspects of AFE, emphasizing the importance of considering macroscopic and microscopic findings in diagnosing and understanding this complex obstetric emergency.

History and Physical

The medical history or current health details of a patient experiencing an AFE may reveal factors such as advanced maternal age, multiple pregnancies, placenta-related issues (accreta, abruption, previa), preeclampsia, gestational diabetes, polyhydramnios, amniocentesis, use of amnioinfusion, amniotomy, cervical lacerations, or any surgery on the gravid uterus.[16] In the classic scenario, women in the later stages of labor suddenly develop acute shortness of breath accompanied by hypotension. Preceding other symptoms, there may be signs of agitation, anxiety, altered mental status, or a sense of impending doom. Seizures may ensue, leading to cardiac arrest, followed by massive hemorrhage associated with DIC, ultimately resulting in death—often within an hour of onset. Statistics indicate that 53% of females with AFE present at or just before delivery, while the remainder present, on average, 19 minutes after delivery.[16]

AFE commonly manifests with cardiac arrest, but other presentations include respiratory collapse and DIC. Many patients lose consciousness, and some may exhibit seizure-like activity (10% to 50%), likely due to brain anoxia.[16] The physical examination typically reveals a patient in cardiovascular collapse, marked by severe hypoxemia, hypotension, and cyanosis. The classic triad of AFE consists of hypoxia, hypotension, and coagulopathy, with a normal body temperature.[3] Funduscopic examination may detect minute bubbles in retinal arteries. Tachypnea may be present, often accompanied by the characteristic holosystolic high-pitched murmur of tricuspid regurgitation. This murmur is loudest at the lower left sternal border, radiating to the right sternal edge. Hemorrhage can range from massive to minimal, and uterine atony (83%) exacerbates bleeding. Initial bleeding typically occurs from the vagina but may also be observed in surgical incisions. Full-blown DIC is observed in approximately 83% of patients. Premonitory symptoms, such as shortness of breath or agitation, may precede cardiovascular collapse.[16]

Evaluation

The diagnosis of AFE relies on exclusionary criteria after the occurrence of a clinical scenario that aligns with its characteristics.[17] It is fundamentally a clinical diagnosis, as no reliable, definitive test for AFE exists. The suspicion of AFE arises when sudden dyspnea, dysphoria, hypotension, cardiovascular collapse, and coagulopathy manifest following actions during the peripartum period, such as active labor, rupture of membranes, vaginal delivery, or cesarean section. AFE has also been observed during or after elective pregnancy terminations, whether induced or surgical. Initial evaluation typically occurs during aggressive cardiopulmonary resuscitation, focusing on the 2 main system failures: hemodynamic and hematologic.[17]

Transthoracic echocardiography (TTE) or transesophageal echocardiography (TEE) plays a crucial role in diagnosis when available. TEE is preferred if patient stability is achieved. Significant echocardiographic findings in AFE include right ventricle dilatation, hypokinesis, overload, tricuspid regurgitation, and right atrial enlargement.[17] Early cardiac thrombi may be identified in the enlarged right ventricle or right atrium.[7] A characteristic feature associated with AFE is the bowing of the intraventricular septum into the left ventricle, leading to left ventricular obstruction and systolic dysfunction, resembling the shape of the letter 'D.'[17]

Immediate blood collection is essential for an urgent type and crossmatch, complete blood count, comprehensive metabolic panel, and a full coagulation panel, encompassing platelets, prothrombin time, partial thromboplastin time, bleeding time, fibrinogen, d-dimer, and fibrin degradation products (FDPs).[17] The International Society on Thrombosis and Hemostasis (ISTH) provides a formal scoring system for determining DIC presence in pregnancy based on platelet count, international neutralization ratio (INR), and fibrinogen level. Scores >3 indicate the presence of DIC in pregnancy.[18][17]

Establishing precise criteria for diagnosing AFE has been challenging due to the absence of a single definitive test. Various international standards have been introduced to define AFE, with the American Society for Maternal-Fetal

Medicine (SMFM) establishing objective criteria following a consensus symposium with the Amniotic Fluid Embolism Foundation in 2016.[18] The criteria stipulate the presence of the following conditions:

1. Sudden cardiopulmonary collapse or hypotension (systolic blood pressure <90 mmHg) with hypoxia (SpO2 <90%).
2. Severe hemorrhage or DIC according to the ISTH definition.
3. Symptomatology occurs either during labor or placental delivery (or up to 30 minutes later).
4. Absence of fever or other explanations for the observed findings.[13][19]

The SMFM acknowledges that there may be cases falling outside these parameters, such as during pregnancy terminations. They clarify that their primary objective is to establish standardized criteria for research reporting. While recognizing that their standards may encompass numerous outlier cases, they hope to minimize such occurrences.[19] The critical clinical findings associated with AFE include coagulopathy, pulmonary hypertension, and neurologic symptoms. Some authors have proposed a modified version of the above definition that would consist of premonitory signs, like seizures, agitation, anxiety, feelings of imminent death, confusion, and fainting. It has been recommended that the diagnostic criteria proposed by SMFM be further validated in future large prospective cohort studies.[18]

Treatment / Management

Due to the rarity of AFE, many obstetric clinicians lack experience in managing such cases. To mitigate panic and confusion and enhance coordination among healthcare providers, it has been proposed that a checklist could be a beneficial cognitive aid in the initial and immediate management of a patient with an AFE.[20] The overall treatment approach for a patient with an AFE is supportive, with initial management aligning with the "ABC" principles, prioritizing the support of the airway, breathing, and circulation.[20]

Prompt and effective cardiopulmonary resuscitation is the cornerstone of managing an acute AFE. Chest compressions should be initiated immediately, providing 100 to 120 compressions per minute with 1 ventilation every 6 seconds. It is crucial to avoid excessive ventilation, as this can lead to a decrease in cardiac output.[20]

During chest compressions, defibrillator pads should be applied without interruption, and a non-synchronized shock should be administered if the cardiac rhythm is shockable. Cardiopulmonary resuscitation should continue for 2 minutes, followed by a pause to check for a pulse and analyze the rhythm. It is recommended to change providers, ensuring that no one person gives chest compressions for more than 2 minutes.[20] If indicated, a second shock may be administered, and the process should be repeated.

In the absence of intravenous (IV) access, it is advisable to establish an interosseous line in the humeral head for fluid and medication administration. If the cardiac rhythm persists, the administration of epinephrine at a dose of 1 mg every 3 to 5 minutes should be initiated through either the IV or interosseous line.[20] In the presence of cardiac arrest with no return of circulation within approximately 4 minutes, preparations for delivery should be made. This can involve an operative delivery or perimortem cesarean delivery if the fetus is at a gestational age considered viable.[20]

The release of vasoconstrictors often results in increased pulmonary vascular resistance, leading to right ventricular failure. TTE findings can help recognize this failure. Managing right ventricular failure involves adjusting ventilator settings, avoiding fluid overload, and using medications such as norepinephrine for blood pressure control. Preferred inotropic support includes dobutamine or milrinone, and vasodilation with epoprostenol is recommended.[20]

Coagulopathy induced by tissue factor activation, primarily through Factor VII activation, should be addressed with a 1:1:1 ratio of packed red blood cells, platelets, and fresh frozen plasma to maintain fibrinogen levels above 150 mg/dL to 200 mg/dL.[20] Tranexamic acid should be given promptly, as this antifibrinolytic is readily available and safe to use with obstetrical hemorrhage.[21] Treating with large fluid volumes should be avoided. If prolonged CPR of 10 minutes or longer is needed, or if right ventricular failure is unresponsive to medical management, consideration may be given to extracorporeal membrane oxygenation (ECMO). Anticoagulation treatment may be necessary to decrease clotting activation, platelet consumption, and the consumption of coagulation factors.[20]

The prompt evacuation of the fetus, termed "resuscitative hysterotomy," is a pivotal component of AFE treatment, distinct from perimortem cesarean section, with the primary aim of improving maternal hemodynamics. Given its time-sensitive nature, this procedure is likely to be conducted at the site of the AFE, as transport to an operating room within the optimal 1- to 2-minute window may not be feasible.[20]

For the mother, comprehensive care involves securing the airway, effective ventilation, appropriate fluid management, and judicious use of vasopressors. Intra-arterial lines facilitate real-time pressure measurement and frequent arterial blood gas sampling. A central venous pressure line aids in assessing right-sided preload. Recognizing the adverse impact of copious fluid administration on clotting factors and bleeding, the decision to transition to vasopressors should be contemplated earlier as compared to later when resuscitating a patient with significant bleeding from other causes.[20]

Pulmonary hypertension and right heart failure commonly accompany AFE, making inotropes, pulmonary vasodilators, and after-load reducing agents paramount to treatment.[20] Vasopressor support should be initiated with norepinephrine if needed. Dobutamine and milrinone, as inotropes, provide pulmonary vasculature dilation and a decrease in right ventricular afterload. Epoprostenol, whether inhaled or intravenous, for pulmonary vasodilation may be used instead of inhaled nitric oxide or sildenafil with similar outcomes for right ventricular failure.[20] Ideal management involves maintaining a mean arterial pressure (MAP) >65 mmHg, a cardiac index >2 L per meter squared, an adequate urine output of 40 mL/hr to 50 mL/hr, and a PaO2/FiO2 ratio >250.

Extracorporeal membrane oxygenation (ECMO) life support has proven successful for refractory cardiogenic shock secondary to AFE when severe right ventricular dysfunction does not respond to medical management.[20] ECMO provides respiratory and hemodynamic support through the femoral vasculature until the right ventricle's function improves. Any patient persisting in cardiopulmonary collapse should have femoral arterial and venous 4 Fr sheaths placed in anticipation of ECMO. Transferring to tertiary facilities capable of ECMO may require early decision-making and awareness of local capabilities.[7] Anticoagulation-free ECMO should be considered if ongoing bleeding or DIC is present.[20]

Following fetal delivery, managing hemorrhage, uterine atony, and the resulting coagulopathy (DIC) is the next step.[21] Empirical administration of packed red blood cells (pRBCs), fresh frozen plasma (FFP), and platelets in a 1:1:1 ratio has been traditional. However, cryoprecipitate is preferred over FFP to reduce volume overload, given its concentrated clotting factors, including factor VIII, von Willebrand factor, and fibrinogen. Tranexamic acid may be administered for fibrinolysis.[22] Blood products are used for fluid resuscitation to mitigate the risk of volume overload.[20]

Standard laboratory turnaround times for clotting parameters may be too slow for actively bleeding patients. Viscoelastic hemostatic assay (VHA)-guided algorithms have demonstrated reduced transfusion requirements, offering improved outcomes. Whole blood viscoelastic hemostatic assays, such as thromboelastography and rotational thromboelastometry, performed at the bedside, provide minute-to-minute evaluation of clinically relevant information, aiding in the rapid assessment and treatment of major obstetric hemorrhage, as seen in AFE. These measurements include fibrinogen levels, platelet count and function, and evaluation of the entire extrinsic clotting pathway, guiding appropriate administration of cryoprecipitate, fibrinogen, prothrombin complex concentrates, platelets, FFP, and pRBCs.[20]

While investigational, anecdotal reports in the literature cite the use of several other medications, including rivaroxaban, a factor Xa inhibitor, C1 esterase inhibitor concentrate, ketorolac, ondansetron, and aminocaproic acid.[20]

Obstetrical management must prioritize the rapid evacuation of the fetus, usually by cesarean section. The Society for Maternal-Fetal Medicine recommends this for all fetuses over 23 weeks gestational age. Evacuation of the uterus may not improve the clinical situation with a previable fetus or even with a gestational age of less than 20 weeks. Ongoing resuscitation of the mother in the delivery room, with an anesthesiologist or critical care provider overseeing cardiovascular resuscitation, must continue during the infant's extraction. Shifting the gravid uterus to the left is recommended, relieving aortocaval compression.[20]

This multidisciplinary team, capable of initiating neonatal resuscitation, should include a neonatologist, given that the majority of these infants are born with a low Apgar score.The obstetrician may opt for several procedures to alleviate ongoing uterine hemorrhage, such as uterine artery ligation or embolization. Circumferential B-Lynch, Hayman, or Pereira compression sutures have been used to compress the atonic uterus and staunch bleeding. However, in the setting of massive hemorrhage and an atonic uterus, emergency hysterectomy may be the best course, required in approximately 50% of patients with severe and ongoing coagulopathy.[20]

Debriefing is recommended to review processes, identify areas for improvement, and offer emotional support to the patient, family, and medical providers. Reporting all cases of AFE to the international AFE registry contributes to improving outcomes in future situations. Simulations and drills are valuable in preparing the care team for these unusual yet emergent clinical situations.[20]

Differential Diagnosis

AFE is prone to misdiagnosis, prompting ongoing efforts to clarify its etiology, risk factors, and pathogenesis.[8] Vigilance must be taken for potentially reversible causes, including hypovolemia, hypoxemia, and hypothermia. Consideration of underlying reversible diagnoses such as myocardial infarction or tamponade, acidosis, hyperkalemia, and tension pneumothorax is warranted.[20] The differential diagnosis for a pregnant patient experiencing complete cardiovascular collapse during or around the time of delivery, followed by significant hemorrhage, should encompass the following: [4]

- Pulmonary embolism (PE)
- Peripartum cardiomyopathy
- Septic shock
- Aortic dissection
- Magnesium toxicity [20]
- Air or cholesterol embolism
- Myocardial infarction
- Venous air embolism
- Eclampsia
- Aspiration
- Toxic reaction to anesthetic medications
- Anaphylaxis
- Obstetrical hemorrhage causing coagulopathy and shock
- Cephalad spread of spinal anesthetic [20]

AFE exhibits some similarities to pulmonary embolus but lacks an ongoing coagulopathy seen in PE. Postpartum cardiomyopathy would likely manifest significant ST-T wave changes on electrocardiography, accompanied by predominant symptoms of left-sided congestive heart failure. Bedside echocardiography (TEE or TTE) can aid in distinguishing AFE with its classic right ventricular dilatation, overload pattern, and septal bowing into the left ventricle.

Septic shock typically presents with the classic systemic inflammatory response syndrome (SIRS) picture and is unlikely to lead to sudden cardiovascular collapse. Myocardial infarction, unless antecedent to the cardiac arrest, would display typical ST-T wave changes and elevated serial cardiac enzymes, observable through bedside echocardiography. Venous air embolism usually presents with wheezing, gasping, and chest pain before cardiovascular collapse. Eclampsia may be suggested by hypertension, edema, proteinuria, headaches, or seizures preceding the collapse. Anaphylaxis should exhibit premonitory symptoms like wheezing, dyspnea, rash, urticaria,

and a period of hypotension before cardiovascular decompensation. Cephalad distribution of spinal anesthetic would present with an elevated sensory level, weakness of the upper extremities, difficulty in speaking, dysphagia, and bradycardia.[20]

Prognosis

AFE stands as a prominent cause of maternal mortality in developed countries, with a historical mortality rate initially reported at 61%, but recent data suggests a lower rate of around 10%.[20] Early and effective management of cardiac arrest significantly enhances survival, with a case fatality rate ranging between 11% to 26% in developed nations.[20]

Tragically, within the first hour following an AFE, an estimated 50% of patients succumb, and two-thirds face mortality within 5 hours. The peak period of death has been noted to be 1 to 12 hours after the AFE occurs.[3] A California-based study indicated that 26.4% of affected pregnant patients died, while 66% developed DIC.[20] Maternal survival remains uncommon, but prompt recognition and resuscitation improve prognoses. The United Kingdom AFE registry reported a 37% mortality rate, with 7% of survivors experiencing neurological impairment.[20]

Survivors of AFE often grapple with substantial neurological, pulmonary, and cardiovascular deficits, affecting two-thirds of these recently pregnant patients. Recurrence risks are uncertain, but instances of successful subsequent pregnancies have been documented.[20] Elective cesarean delivery recommendations for future pregnancies to mitigate labor-related risks are contentious.

Infant mortality rates hover around 30%, accompanied by elevated risks of hypoxic-ischemic encephalopathy, cerebral palsy, and cognitive disabilities among survivors. The grim statistics extend to stillbirth and neonatal death, reaching rates as high as 10% to 40%. Patients who survive pregnancies complicated by AFE commonly grapple with depression and post-traumatic stress disorder (PTSD).[20]

While initial rates of neurologically intact survival were reported at 15%, recent data suggests an improvement, with estimates nearing 46%. Advances in diagnosis, medical management, and a deeper understanding of AFE's pathophysiology contribute to enhanced survival rates. Early recognition has not only benefited maternal outcomes but has also resulted in improved neonatal mortality and morbidity rates.[20]

Complications

Survivors of AFE may contend with a spectrum of significant complications, including the following:

- Renal failure
- Cardiac failure
- Prolonged respiratory failure leading to adult respiratory distress
- Myocardial infarction
- Arrhythmias
- Cardiomyopathy
- Congestive heart failure
- Left ventricular systolic dysfunction
- Prolonged coagulopathy
- Respiratory failure (extended)
- Prolonged bronchospasm
- Liver failure
- Cardiogenic pulmonary edema
- Seizures

- Anoxic encephalopathy
- Various cognitive or neurologic impairments

Infants delivered emergently during maternal AFE are at heightened risk for sustaining hypoxic-ischemic encephalopathy (HIE). This often results in a significantly cognitively impaired child, potentially manifesting chronic epilepsy, motor impairment, and developmental delay.[20][3]

Deterrence and Patient Education

AFE is an exceptionally severe condition characterized by its sudden onset, and unfortunately, its occurrence is largely unpredictable. Despite its unforeseeable nature, there are limited preventive measures that can be taken. To minimize the risk of AFE, care should be exercised during certain maneuvers, such as the insertion of a pressure catheter and intraamniotic infusion therapy. Additionally, efforts should be made to refrain from incising the placenta during cesarean delivery whenever possible.[15]

Future studies are crucial to identifying individuals at the highest risk, as long-term mental and physical health consequences are prevalent in these patients. It is noteworthy that only around 60% of women affected by AFE manage to return to their previous state of well-being. Further research can contribute to a better understanding of risk factors and improve preventive strategies.

Enhancing Healthcare Team Outcomes

The abrupt onset of cardiovascular collapse and profound coagulopathy in mothers, coupled with the imperative for neonatal resuscitation, renders the management of AFE highly intricate and demanding for any interdisciplinary team. Successful outcomes for both the mother and infant hinge on seamless coordination among various healthcare professionals, including obstetricians, maternal-fetal specialists, anesthesiologists, labor and delivery nurses, neonatologists, intensivists, hematologists, perfusionists, respiratory therapists, and neonatal intensive care unit nurses.

A multidisciplinary approach, facilitated within a high-acuity level hospital setting, is crucial for optimizing maternal and infant survival outcomes in cases of AFE.[15] This collaborative effort ensures that expertise from diverse medical specialties converges to address the multifaceted challenges posed by AFE, thereby enhancing the overall quality of care and response to this challenging obstetric emergency.

Review Questions

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